25901550 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSESTA200 - DATA ANALYSIS USING JAMOVI

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **3** marks

1. Compare the roles of the Data Tab and Analyses Tab. [U] / [CO1]
2. Explain the "omv" file format and its importance in data sharing. [U] / [CO1]
3. What is the primary purpose of the Rj Editor module in JAMOVI? [K] / [CO1]
4. State any three limitations of JAMOVI compared to R. [U] / [CO1]
5. Define partial correlation. [K] / [CO2]
6. Define Q-Q plot. [K] / [CO2]
7. Define rate ratio. [K] / [CO2]
8. What is scatterplot matrix? [K] / [CO2]
9. Give one real-life example where two-way ANOVA can be applied. [U] / [CO4]
10. Which graphical methods are commonly used to assess normality? [K] / [CO4]

11. Briefly explain principal component analysis. [U] / [CO5]

12. Define McDonald's Omega in reliability analysis. [K] / [CO5]

[3x10 = 30]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **7** marks

13. Explain the step-by-step process of installing a new module from the JAMOVI library. [U] / [CO1]

14. List and briefly explain four key functions found within the main "Hamburger" menu. [K] / [CO1]

15. (a) Generate 1000 observations from Uniform (0,1). (b) Draw histogram. (c) Take repeated samples (size=30) and compute means. (d) Plot sampling distribution of means. [A] / [CO1]

16. Explain the steps to create a new computed variable in JAMOVI. [U] / [CO1]

17. The heights of 10 plants of a given garden are found to be 70,67,62,68,61,68,70,64,64,66 cms. Is it reasonable to believe that the average height is greater than 64 cms? Test, at the 5% level of significance with 9 degrees of freedom, [A] / [CO4]

18. Describe the applications of reliability analysis. [U] / [CO5]

[7x4 = 28]

Part C

Essay Questions

Answer any **1** questions

Each question carries **12** marks

19. a) What is effect size? Explain Cohen's d in the context of independent sample t-test. (b) Below are given the gain in weights (in lbs) of dogs fed on two diets A and B.
    Diet A : 25,32,30,34,24,14,32,24,30,31,35,25.
    Diet B : 44,34,22,10,47,31,41,30,32,35,18,21.
    In JAMOVI, test whether the two diets differ significantly and interpret the results. [A] / [CO4]

20. Explain predictive measures in logistic regression. [U] / [CO3]

[12x1 = 12]